

B.TECH 1ST YEAR · ENGINEERING CHEMISTRY

Instrumental Methods and Applications

Spectroscopy · Chromatography · HPLC

UNIT – V

Electromagnetic Spectrum

Beer-Lambert Law

UV-Visible Spectroscopy

IR Spectroscopy

Chromatography

HPLC

40+ Solved Numericals

Q&A Bank

✦ Conceptual depth from zero level · Exam-oriented diagrams · Step-by-step solutions

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How to use this module: Each section builds on the previous. For exam preparation, focus on formula boxes, key-point boxes, and solved problems. Diagrams are exam-oriented and should be practiced for 5-mark answers.

SECTION 1

Introduction to Instrumental Methods

1.1 Why Do We Need Instrumental Analysis?

BASIC INTUITION

Imagine you have a coloured solution and you want to know: *How much dye is dissolved in it?* You could taste it (dangerous!), or you could shine light through it and measure how much light is absorbed. The more dye, the more light absorbed. This is the essence of **spectroscopic instrumental analysis**.

Classical chemical analysis (gravimetry, titrimetry) involves weighing, precipitating, and filtering — slow and labour-intensive. Modern industry needs **rapid, accurate, non-destructive** analysis of samples. Instrumental methods fulfil this need by exploiting the interaction of *energy* (light, electricity, heat) with *matter*.

1.2 Classification of Instrumental Methods

Category	Property Measured	Examples
Spectroscopic Methods	Absorption / emission of EMR	UV-Vis, IR, NMR, AAS
Electroanalytical	Electrical potential / current	Potentiometry, Coulometry
Chromatographic	Differential migration	TLC, HPLC, GC
Thermal Methods	Heat changes	DSC, TGA

1.3 Scope of Unit V

This unit focuses primarily on **spectroscopic methods** (UV-Vis and IR) and **chromatographic methods** (TLC and HPLC). These are the backbone of quality control in pharmaceutical, food, petrochemical, and environmental industries.

KEY POINTS TO REMEMBER

- Instrumental methods are faster, more sensitive, and often non-destructive.
- Spectroscopy = study of interaction of electromagnetic radiation (EMR) with matter.
- Chromatography = separation of mixtures based on differential movement through stationary and mobile phases.
- HPLC = high-pressure version of liquid chromatography with superior resolution.

SECTION 2

Electromagnetic Spectrum

2.1 What is Electromagnetic Radiation?

STEP 1 – BASIC IDEA

When you feel warmth from the sun, see colours, get an X-ray at a hospital, or use a microwave oven — all of these involve **electromagnetic radiation (EMR)**. EMR is energy that travels through space as oscillating electric and magnetic fields, perpendicular to each other and to the direction of propagation.

STEP 2 – WAVE PROPERTIES

EMR behaves as a wave. The key properties are:

- **Wavelength (λ)** — distance between two successive crests (unit: meters, nm, cm^{-1})
- **Frequency (ν)** — number of oscillations per second (unit: $\text{Hz} = \text{s}^{-1}$)
- **Speed (c)** — in vacuum, $c = 3 \times 10^8 \text{ m/s}$ for all EMR

2.2 Key Relationships

FUNDAMENTAL EQUATIONS OF ELECTROMAGNETIC RADIATION

$$c = \lambda \nu \quad (\text{Wave equation})$$

$$E = h\nu = hc/\lambda \quad (\text{Planck's equation})$$

$$\tilde{\nu} = 1/\lambda \quad (\text{Wavenumber in } \text{cm}^{-1})$$

Symbol Legend: c = speed of light ($3 \times 10^8 \text{ m/s}$) | λ = wavelength (m or nm) | ν = frequency (Hz) | h = Planck's constant ($6.626 \times 10^{-34} \text{ J}\cdot\text{s}$) | E = photon energy (J) | $\tilde{\nu}$ = wavenumber (cm^{-1})

Important insight: Higher frequency $\rightarrow$ shorter wavelength $\rightarrow$ *higher energy*. Gamma rays (very high ν) carry far more energy than radio waves (very low ν). This determines which molecular events each region can trigger.

2.3 The Electromagnetic Spectrum – Full Chart

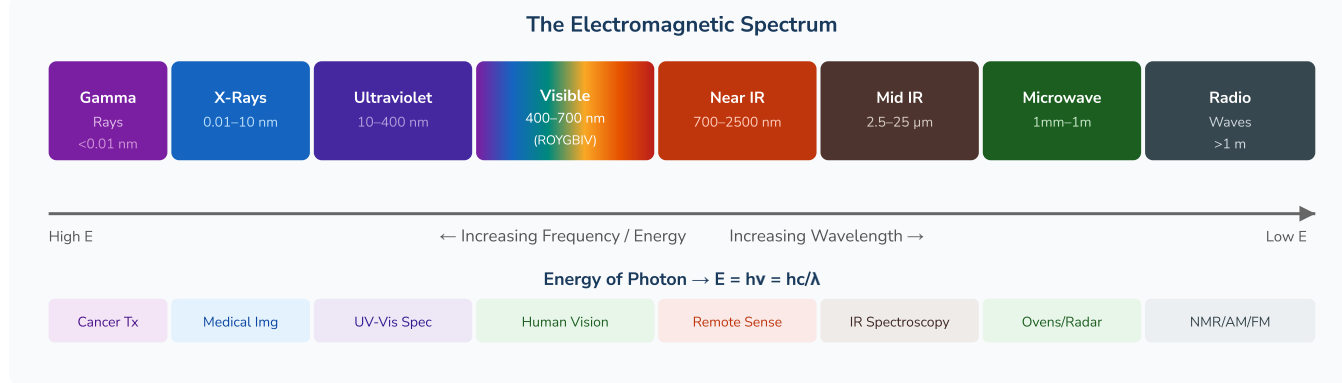


Fig 2.1 – The Complete Electromagnetic Spectrum with wavelength ranges and applications

2.4 Regions Most Relevant to This Unit

Region	Wavelength Range	Frequency (Hz)	Molecular Event
UV	10–400 nm	10^{15} – 10^{16}	Electronic transitions (inner)
Visible	400–700 nm	4×10^{14} – 7×10^{14}	Electronic transitions (outer)
Near IR	700–2500 nm	10^{13} – 10^{14}	Overtone vibrations
Mid IR	2.5–25 μ m	10^{12} – 10^{13}	Fundamental vibrations
Microwave	1mm–1m	10^9 – 10^{12}	Rotational transitions

KEY POINTS – ELECTROMAGNETIC SPECTRUM

- All EMR travels at speed $c = 3 \times 10^8$ m/s in vacuum.
- Energy is directly proportional to frequency: $E = h\nu$.
- Energy is inversely proportional to wavelength: $E = hc/\lambda$.
- UV-Vis region causes electronic transitions; IR region causes vibrational transitions.
- Wavenumber ($\tilde{\nu} = 1/\lambda$ in cm^{-1}) is widely used in IR spectroscopy.

SECTION 3

Interaction of Radiation with Matter

3.1 What Happens When Light Hits a Molecule?

CONCEPTUAL STARTING POINT

When a beam of electromagnetic radiation falls on a substance, three things can happen:

- **Transmission** – radiation passes through unchanged (transparent material)
- **Reflection** – radiation bounces back
- **Absorption** – radiation energy is taken up by the molecule → *this is what we measure in spectroscopy*

For absorption to occur, the energy of the photon must exactly match the energy gap between two quantum states of the molecule (ground state → excited state).

3.2 Quantum View of Absorption

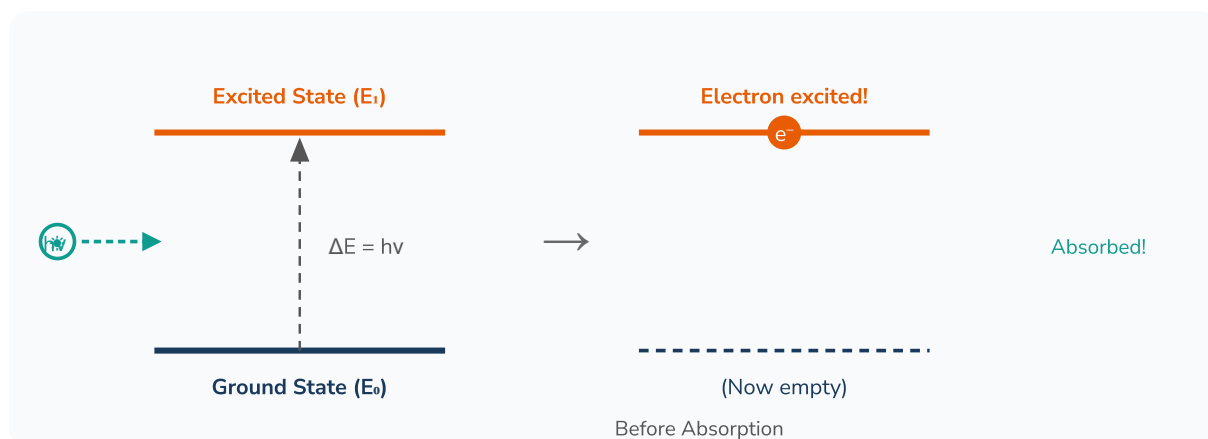


Fig 3.1 – Absorption of a photon promotes an electron from ground state to excited state when $E = h\nu = \Delta E$

3.3 Types of Energy Transitions

Type of Transition	Energy Required	Spectral Region	Example
Nuclear spin flip	Very low	Radio wave	NMR spectroscopy
Molecular rotation	Low	Microwave	Rotational spectra
Molecular vibration	Moderate	IR	IR spectroscopy
Electronic (outer)	High	UV-Visible	UV-Vis spectroscopy
Electronic (inner)	Very high	X-Ray	X-Ray absorption

3.4 Transmittance and Absorbance

When light of intensity I_0 enters a sample and intensity I exits:

Transmittance: $T = I / I_0$ (dimensionless, 0 to 1)

$$\%T = (I / I_0) \times 100$$

Absorbance: $A = \log_{10}(I_0 / I) = \log_{10}(1/T) = -\log_{10}(T)$

Intuition: If $T = 1$ (100%), no light is absorbed $\rightarrow A = 0$. If $T = 0.1$ (10%), $A = \log(10/1) = 1$. Higher absorbance means the sample absorbs more light.

SECTION 4

Absorption of Radiation & Beer-Lambert Law

4.1 Building the Beer-Lambert Law from Scratch

STEP 1 – LAMBERT'S CONTRIBUTION (1760)

Johann Lambert observed: *"For a fixed concentration, the fraction of light absorbed is proportional to the path length (l) through the sample."*

Mathematically: $-dI/dl = k_1 \times I \rightarrow$ After integration: $\log(I_0/I) \propto l$

STEP 2 – BEER'S CONTRIBUTION (1852)

August Beer observed: *"For a fixed path length, the fraction of light absorbed is proportional to the concentration (c) of the absorbing species."*

Mathematically: $\log(I_0/I) \propto c$

STEP 3 – COMBINED BEER-LAMBERT LAW

Combining both observations:

BEER-LAMBERT LAW

$$A = \epsilon \cdot c \cdot l$$

$$\log(I_0/I) = \epsilon \cdot c \cdot l$$

Symbol Legend:

A = Absorbance (dimensionless) | ϵ = Molar absorptivity / molar extinction coefficient ($\text{L}\cdot\text{mol}^{-1}\cdot\text{cm}^{-1}$) | c = Concentration (mol/L or M) | l = Path length (cm)

4.2 Graphical Representation

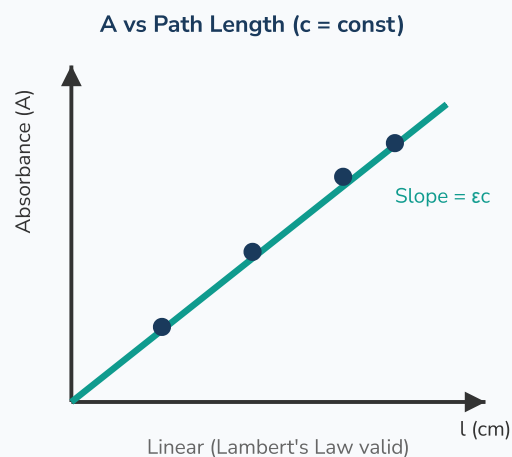
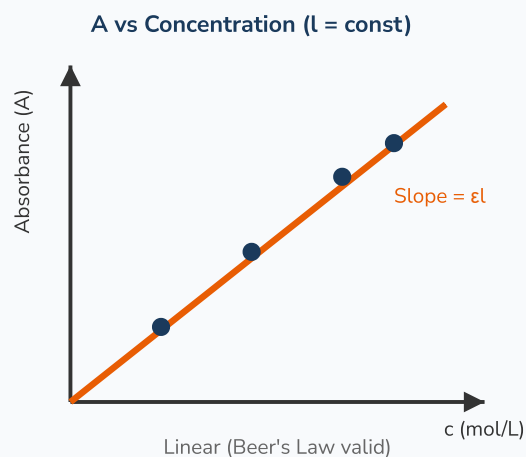


Fig 4.1 – Beer-Lambert Law: A varies linearly with both concentration and path length

4.3 Physical Meaning of Molar Absorptivity (ϵ)

ϵ is a measure of how strongly the molecule absorbs light at a specific wavelength. It is a characteristic of the molecule — not the instrument or concentration. A high ϵ means the molecule is a strong absorber.

- $\epsilon < 100$: Very weak absorber
- $100 < \epsilon < 10,000$: Moderate absorber
- $\epsilon > 10,000$: Very strong absorber (e.g., aromatic compounds)

4.4 Limitations and Deviations from Beer-Lambert Law

IMPORTANT LIMITATIONS (EXAM FAVOURITE!)

Real (Chemical) Deviations:

- At high concentrations (>0.01 M), solute molecules interact with each other, changing ϵ .
- Chemical equilibria (dimerization, ionization) change the absorbing species.

Instrumental Deviations:

- Non-monochromatic radiation: Beer's law strictly holds only for monochromatic light.
- Stray light: Light reaching detector without passing through sample causes errors.
- Fluorescence and scattering also cause deviations.

Physical Deviations:

- High concentration changes refractive index, affecting measured absorbance.

BEER-LAMBERT LAW – MUST-KNOW POINTS

- $A = \epsilon cl$ — the fundamental law of quantitative spectrophotometry.
- Absorbance is additive: for multiple species, $A_{\text{total}} = A_1 + A_2 + \dots$
- Beer's law is valid only at low to moderate concentrations (<0.01 M typically).
- A calibration curve (A vs c) is used to determine unknown concentrations.

SECTION 5

UV-Visible Spectroscopy – Basic Concept

5.1 What is UV-Vis Spectroscopy?

BASIC INTUITION

Why does a tomato look red? Because it absorbs violet, blue, and green light, and reflects red. A colorimeter or UV-Vis spectrophotometer measures *how much of each wavelength is absorbed* by a substance. This tells us what molecules are present and how much of them there is.

UV-Visible spectroscopy involves the absorption of radiation in the **ultraviolet (200–400 nm)** and **visible (400–700 nm)** regions. This absorption causes **electronic transitions** — electrons jump from lower energy orbitals to higher energy orbitals.

5.2 Principle

When UV or visible light passes through a sample:

1. Photons whose energy matches the energy gap (ΔE) between electronic levels are absorbed.
2. The molecule records a characteristic absorption at that wavelength → **absorption spectrum**.
3. The wavelength of maximum absorption is called **λ_{max}** .

$$\Delta E = E_{\text{excited}} - E_{\text{ground}} = h\nu = hc/\lambda$$

The spectrum is plotted as **Absorbance (A) vs Wavelength (λ)**. Each peak corresponds to a specific electronic transition.

5.3 What Information Can We Get?

Parameter	What It Tells Us
λ_{max} (wavelength of max absorption)	Identifies the type of chromophore / compound
ϵ_{max} (molar absorptivity at λ_{max})	Strength of absorption; helps identify compound
Absorbance (A)	Concentration via Beer-Lambert Law
Shape of spectrum	Structural information about the molecule

SECTION 6

Electronic Transitions in UV-Vis Spectroscopy

6.1 Types of Molecular Orbitals

Before understanding transitions, recall the types of electrons in organic molecules:

- **σ electrons** — in sigma bonds (e.g., C–C, C–H); tightly bound
- **π electrons** — in pi bonds (e.g., C=C, C=O); less tightly bound
- **n electrons** — non-bonding lone pair electrons (e.g., on O, N, S, Cl); easily excited

Corresponding antibonding orbitals: σ^* and π^* (no n^*).

6.2 Electronic Transitions – Energy Order

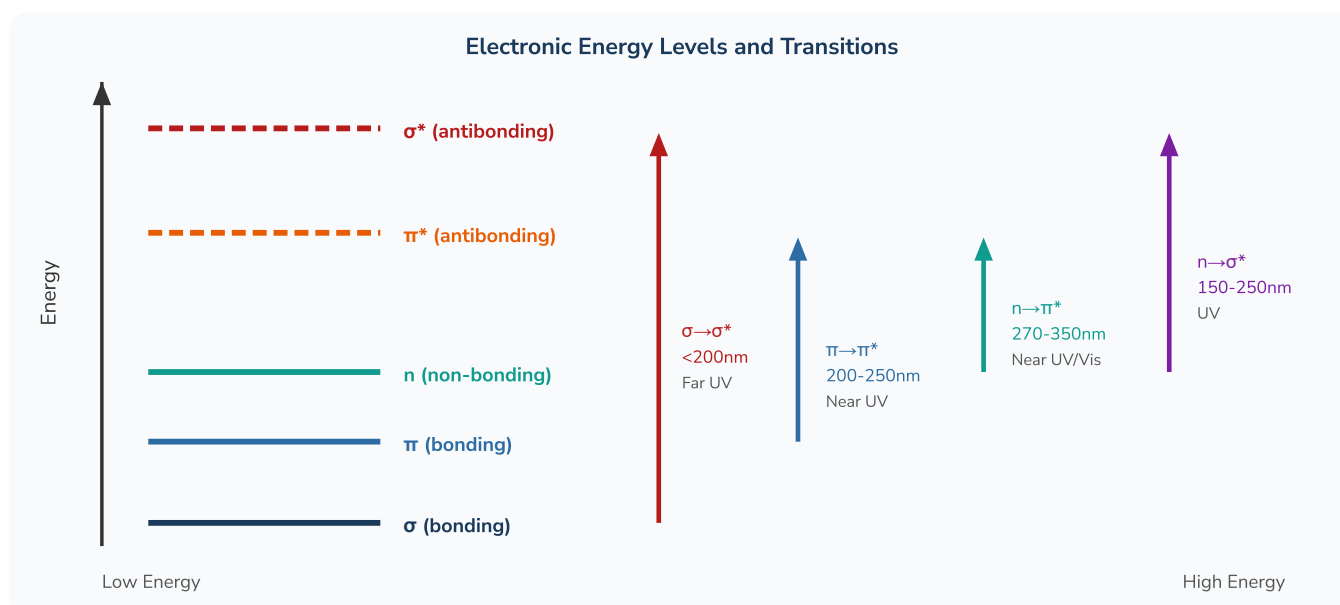


Fig 6.1 – Electronic transitions in UV-Vis spectroscopy (dashed lines = antibonding orbitals)

6.3 Summary of Electronic Transitions

Transition	Wavelength	ϵ	Requirement	Example
$\sigma \rightarrow \sigma^*$	<200 nm (Vacuum UV)	High	C–C, C–H bonds	Alkanes (methane: 125 nm)
$n \rightarrow \sigma^*$	150–250 nm	Low–moderate	Heteroatom lone pairs	Alcohols, ethers, amines
$\pi \rightarrow \pi^*$	170–250 nm	Very high (10^4)	C=C, C=O, aromatic	Benzene 254 nm, ethylene 165 nm
$n \rightarrow \pi^*$	270–350 nm	Low (10–100)	C=O, C=N with lone pair	Acetone: 270 nm ($n \rightarrow \pi^*$)

Memory Aid: In order of increasing energy (decreasing wavelength): $n \rightarrow \pi^* < \pi \rightarrow \pi^* < n \rightarrow \sigma^* < \sigma \rightarrow \sigma^*$

6.4 Bathochromic and Hypsochromic Shifts

- **Bathochromic shift (Red shift)** — λ_{max} moves to longer wavelength (lower energy). Caused by conjugation, auxochromes, polar solvents (for $n \rightarrow \pi^*$).
- **Hypsochromic shift (Blue shift)** — λ_{max} moves to shorter wavelength. Caused by electron-withdrawing groups, loss of conjugation.
- **Hyperchromic effect** — increase in ϵ (more intense absorption).
- **Hypochromic effect** — decrease in ϵ .

SECTION 7

Chromophores and Auxochromes

7.1 What is a Chromophore?

ETYMOLOGY & BASIC IDEA

The word "chromophore" comes from Greek: *chroma* = colour, *phoros* = bearing. A chromophore is any group in a molecule that is responsible for absorbing UV or visible light, thereby causing the molecule to appear coloured or absorb in the UV range.

Definition: A chromophore is a covalently unsaturated group or functional group that gives rise to an electronic absorption band in the UV-Visible region. It usually contains π electrons or non-bonding electrons that can undergo $\pi \rightarrow \pi^*$ or $n \rightarrow \pi^*$ transitions.

7.2 Common Chromophores

Chromophore	Structure	λ_{max} (nm)	ϵ	Transition
Ethylene (C=C)	--C=C--	171	~15,000	$\pi \rightarrow \pi^*$
Acetylene (C $\equiv$ C)	$\text{--C}\equiv\text{C--}$	150–170	~10,000	$\pi \rightarrow \pi^*$
Carbonyl (C=O)	--C=O	166 ($\pi \rightarrow \pi^*$), 280 ($n \rightarrow \pi^*$)	High / Low	$\pi \rightarrow \pi^*$, $n \rightarrow \pi^*$
Nitro (--NO_2)	--NO_2	270	~12	$n \rightarrow \pi^*$
Azo (--N=N--)	--N=N--	340	~10	$n \rightarrow \pi^*$
Benzene ring	Aromatic	254	~200	$\pi \rightarrow \pi^*$
Butadiene (conjugated)	--C=C--C=C--	217	21,000	$\pi \rightarrow \pi^*$

Effect of Conjugation: Extending conjugation shifts λ_{max} to longer wavelengths (bathochromic shift) and increases ϵ . Example: Ethylene (171 nm) $\rightarrow$ Butadiene (217 nm) $\rightarrow$ Hexatriene (258 nm).

7.3 What is an Auxochrome?

Definition: An auxochrome is a substituent group attached to a chromophore that, by itself, does not absorb UV-Vis radiation but when attached to a chromophore, it causes a bathochromic shift (red shift) and intensifies absorption (hyperchromic effect).

Auxochromes typically carry lone pairs of electrons (e.g., --OH , --NH_2 , --OR , --SR , --Cl , --Br). These lone pair electrons interact with the π system of the chromophore via resonance, extending conjugation and shifting λ_{max} .

7.4 Chromophore vs Auxochrome — Comparison

Property	Chromophore	Auxochrome
Absorbs UV-Vis alone?	Yes	No (alone)
Contains	π bonds, lone pairs	Lone pairs (–OH, –NH ₂)
Effect when attached	Provides absorption	Shifts and intensifies
Examples	–C=C–, –C=O, –N=N–	–OH, –NH ₂ , –OR, halogens
Type of shift	—	Bathochromic + Hyperchromic

SECTION 8

UV-Visible Spectrophotometer – Instrumentation

8.1 Block Diagram of UV-Vis Spectrophotometer

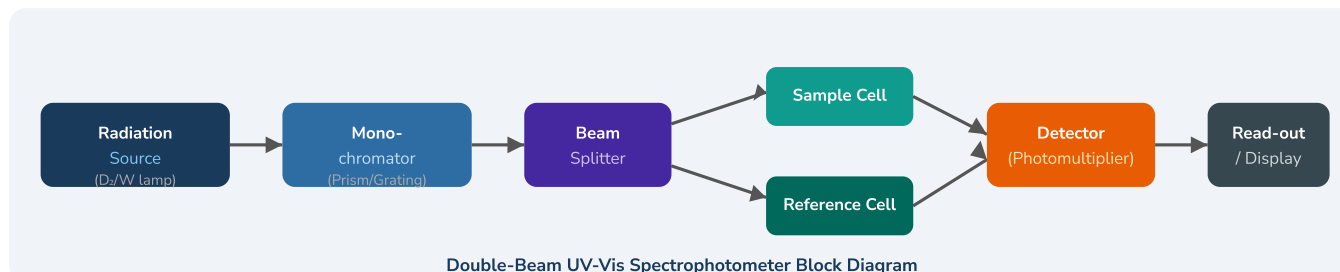


Fig 8.1 – Block diagram of a double-beam UV-Visible spectrophotometer

8.2 Components and Their Functions

Component	Function	Details
Radiation Source	Generates UV/Vis light	Deuterium (D ₂) lamp for UV (190–400 nm); Tungsten-halogen (W) lamp for visible (350–800 nm)
Monochromator	Selects specific wavelength	Uses prism or diffraction grating; isolates narrow band of wavelengths
Beam Splitter	Divides beam into sample and reference	Used in double-beam instruments for simultaneous comparison
Sample Cell (Cuvette)	Holds sample solution	Quartz for UV; glass/plastic for visible; path length = 1 cm standard
Reference Cell	Holds pure solvent (blank)	Compensates for solvent absorption
Detector	Converts light to electrical signal	Photomultiplier tube (PMT) or photodiode array
Readout/Display	Shows absorbance or %T	Digital display, recorder, or computer

8.3 Applications of UV-Visible Spectroscopy

- **Quantitative analysis:** Determination of concentration via Beer-Lambert Law (drugs, vitamins, pollutants)
- **Qualitative analysis:** Identification of compounds by λ_{max} and ϵ
- **Structure elucidation:** Detecting conjugation, chromophores in organic compounds
- **Reaction kinetics:** Following reaction progress by measuring absorbance changes with time
- **Purity determination:** Detecting impurities that absorb at different wavelengths

- **Water/food quality testing:** Colorimetric analysis of metals, nutrients

SECTION 9

Infrared (IR) Spectroscopy – Basic Concept

9.1 What is IR Spectroscopy?

BASIC INTUITION

Imagine a spring connecting two balls (atoms). If you push the balls towards each other and release, they vibrate. Every bond in a molecule acts like a spring — it has a natural vibration frequency. When infrared light of exactly that frequency hits the bond, the bond absorbs the energy and vibrates more intensely. By measuring which IR frequencies are absorbed, we can identify which bonds are present — essentially reading a molecular fingerprint.

9.2 Principle of IR Spectroscopy

IR spectroscopy is based on the absorption of infrared radiation (typically **4000–400 cm^{-1}**) by molecules, causing transitions between vibrational energy levels.

- When IR radiation matches the vibrational frequency of a bond, the bond absorbs that radiation.
- The molecule's **dipole moment must change** during vibration for IR absorption to occur.
- Homonuclear diatomics (H_2 , N_2 , O_2) do NOT absorb IR — they have no dipole change.

$$\text{Wavenumber: } \tilde{\nu} = 1/\lambda \text{ (cm}^{-1}\text{)}$$

$$\text{Hooke's Law (vibration frequency): } \nu = (1/2\pi) \sqrt{k/\mu}$$

$$\text{where } k = \text{force constant (bond strength), } \mu = \text{reduced mass} = (m_1 \cdot m_2) / (m_1 + m_2)$$

Insight from Hooke's Law: Strong bonds (large k) vibrate at higher frequencies. Lighter atoms (small μ) also vibrate at higher frequencies. Thus C–H stretching ($\sim 3000 \text{ cm}^{-1}$) appears at higher wavenumber than C–C stretching ($\sim 1000 \text{ cm}^{-1}$).

9.3 IR Spectral Regions

Region	Wavenumber (cm^{-1})	What It Shows
Functional group region	4000–1500	Characteristic absorptions of specific functional groups
Fingerprint region	1500–400	Unique pattern for each compound; used for identification

The **fingerprint region** is so complex and unique to each molecule that it serves like a fingerprint for identification purposes.

9.4 Key IR Absorption Frequencies

Bond / Group	Wavenumber (cm ⁻¹)	Appearance
O–H stretch (free)	3580–3650	Sharp, strong
O–H stretch (H-bonded)	3200–3550	Broad, strong
N–H stretch	3300–3500	Medium
C–H stretch (alkane)	2850–3000	Medium–strong
C≡C stretch	2100–2260	Weak
C≡N stretch	2200–2260	Medium
C=O stretch (carbonyl)	1680–1780	Very strong, sharp
C=C stretch (aromatic)	1500–1600	Medium
C–O stretch	1000–1300	Strong
C–Cl stretch	600–800	Strong

SECTION 10

Vibrational Modes and Selection Rules

10.1 Types of Molecular Vibrations

Molecules with N atoms have $3N - 6$ vibrational modes (or $3N - 5$ for linear molecules). These fall into two categories:

A. STRETCHING VIBRATIONS

Bond length changes along the bond axis. Can be symmetric (both bonds stretch/compress together) or asymmetric (one stretches while the other compresses).

B. BENDING VIBRATIONS

Bond angle changes while bond lengths remain approximately constant. Types include: scissoring, rocking, wagging, and twisting.

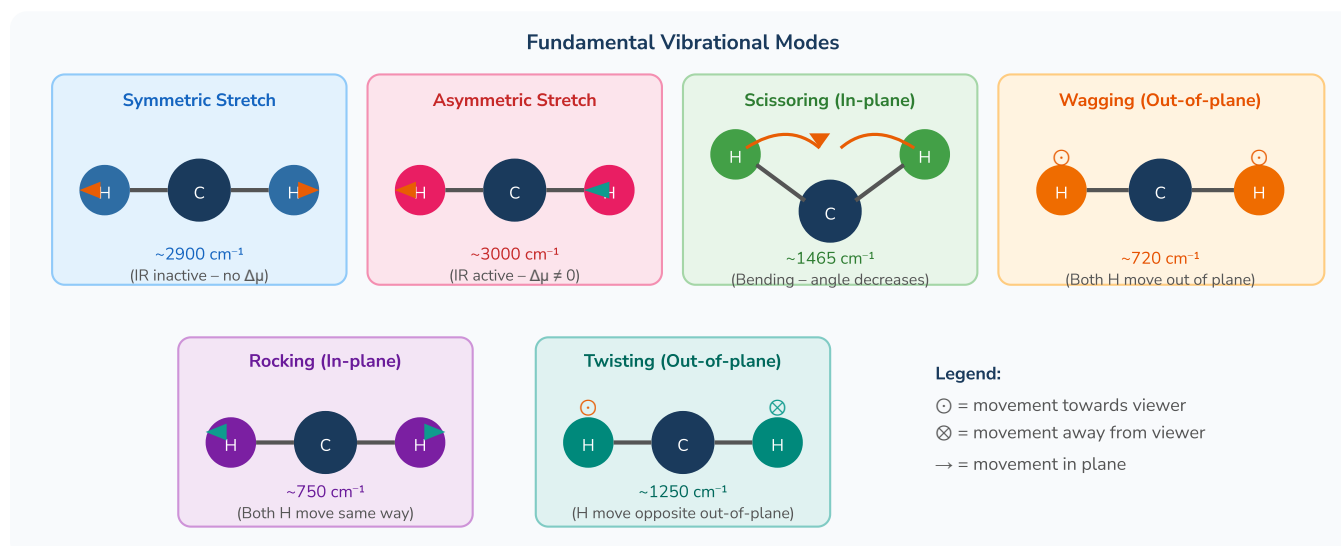


Fig 10.1 – The six fundamental vibrational modes (stretching and bending) of a triatomic molecule

10.2 Selection Rule for IR Activity

IR Selection Rule: A vibrational mode is IR active (will appear in the IR spectrum) *only if* the vibration causes a change in the **dipole moment (μ)** of the molecule.

- Symmetric stretching of symmetrical molecules (like CO_2 symmetric stretch) → **NO dipole change** → **IR inactive**
- Asymmetric stretching of the same molecule → dipole changes → **IR active**
- All vibrations of asymmetric molecules are IR active

10.3 Number of Vibrational Modes

Non-linear molecules: $3N - 6$ vibrational modes

Linear molecules: $3N - 5$ vibrational modes

(N = number of atoms)

Example: Water (H_2O) is non-linear, $N=3 \rightarrow 3(3)-6 = \mathbf{3 \text{ modes}}$. CO_2 is linear, $N=3 \rightarrow 3(3)-5 = \mathbf{4 \text{ modes}}$. Methane CH_4 , $N=5 \rightarrow 3(5)-6 = \mathbf{9 \text{ modes}}$.

SECTION 11

IR Spectrometer – Instrumentation

11.1 Dispersive IR Spectrometer – Block Diagram

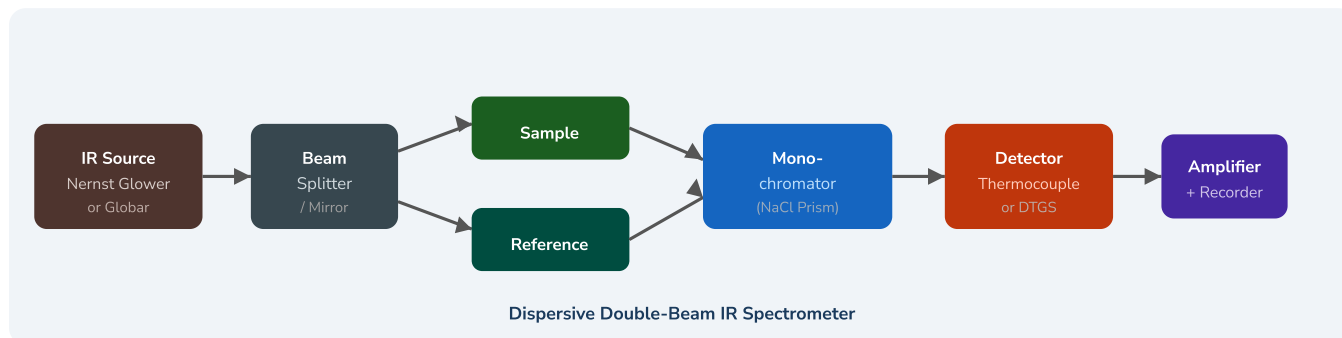


Fig 11.1 – Block diagram of a dispersive double-beam IR spectrometer

11.2 Components of IR Spectrometer

Component	Purpose	Material
IR Source	Emits infrared radiation	Nernst glower ($\text{ZrO}_2/\text{Y}_2\text{O}_3$, 1500–2000 K), Globar (SiC rod), Mercury arc
Sample Cell	Holds sample	KBr or NaCl plates (NOT glass — glass absorbs IR!)
Monochromator	Disperses IR light by wavelength	NaCl or KBr prism; diffraction grating
Detector	Converts IR to electrical signal	Thermocouple, Golay cell, DTGS (pyroelectric detector), MCT
Amplifier/Recorder	Amplifies and records signal	Electronic amplifier + chart recorder / computer

11.3 Sample Preparation for IR

Sample Type	Preparation Method
Solid	KBr pellet (grind sample with KBr, press into transparent pellet); or Nujol mull
Liquid	Thin film between NaCl plates; or in solution in CCl_4/CS_2
Gas	Enclosed in gas cell with long path length

11.4 Applications of IR Spectroscopy

- **Functional group identification:** Identifying O–H, C=O, N–H, etc. in organic compounds
- **Structure determination:** Confirming proposed structures of synthesised compounds

- **Quantitative analysis:** Measuring concentrations using band intensities
- **Polymer identification:** Differentiating polymer types from fingerprint region
- **Quality control:** Checking purity of pharmaceuticals, foods, fuels
- **Forensic science:** Identifying drugs, explosives, fibres

11.5 UV-Vis vs IR Spectroscopy – Comparison

Parameter	UV-Vis Spectroscopy	IR Spectroscopy
Region	200–800 nm	4000–400 cm ⁻¹ (2.5–25 μm)
Type of transition	Electronic	Vibrational
Selection rule	$\Delta E = h\nu$ (quantum)	Change in dipole moment
Sample cells	Quartz (UV); glass (Vis)	KBr, NaCl (NOT glass)
Information	Chromophores, concentration	Functional groups, structure
Chromophore needed	Yes (π , n electrons)	No (any bond that vibrates)
Quantitative use	Excellent (Beer-Lambert)	Moderate
Qualitative use	Limited	Excellent (fingerprint region)

SECTION 12

Chromatography – Principles and Types

12.1 What is Chromatography?

BASIC INTUITION

Imagine dropping a black ink drop on wet filter paper. As water spreads outward, the different dyes in the ink travel at different speeds — red travels farthest, blue stays closer to the centre. Different components of a mixture separate because they interact differently with the paper (stationary phase) and the water (mobile phase). This is chromatography at its simplest!

Definition: Chromatography is a separation technique based on the **differential migration** of components of a mixture between two phases: a *stationary phase* (fixed) and a *mobile phase* (moving). Components with stronger affinity for the stationary phase move slowly; those with weaker affinity move faster — thus separating from each other.

12.2 Key Terms in Chromatography

Term	Meaning
Stationary Phase	Fixed phase; solid or liquid on a support; adsorbs or dissolves components
Mobile Phase	Moving phase; liquid or gas; carries the sample through
Eluent	Solvent used as mobile phase
Eluate	Fractions collected at exit of column
Retention time (t_R)	Time for a component to travel from injection to detector
Resolution (R)	Degree of separation between adjacent peaks
Partition coefficient (K)	$K = \text{Concentration in stationary phase} / \text{Concentration in mobile phase}$

12.3 Classification of Chromatography

Basis of Classification	Type	Mobile Phase	Stationary Phase
Physical state of phases	Gas Chromatography (GC)	Gas	Liquid or solid
	Liquid Chromatography (LC)	Liquid	Solid or liquid
Mechanism	Adsorption Chromatography	Liquid/Gas	Solid (silica, alumina)
	Partition Chromatography	Liquid/Gas	Liquid on support
Operational format	Column Chromatography	Liquid	Solid in tube

12.4 Thin Layer Chromatography (TLC) – Separation Diagram

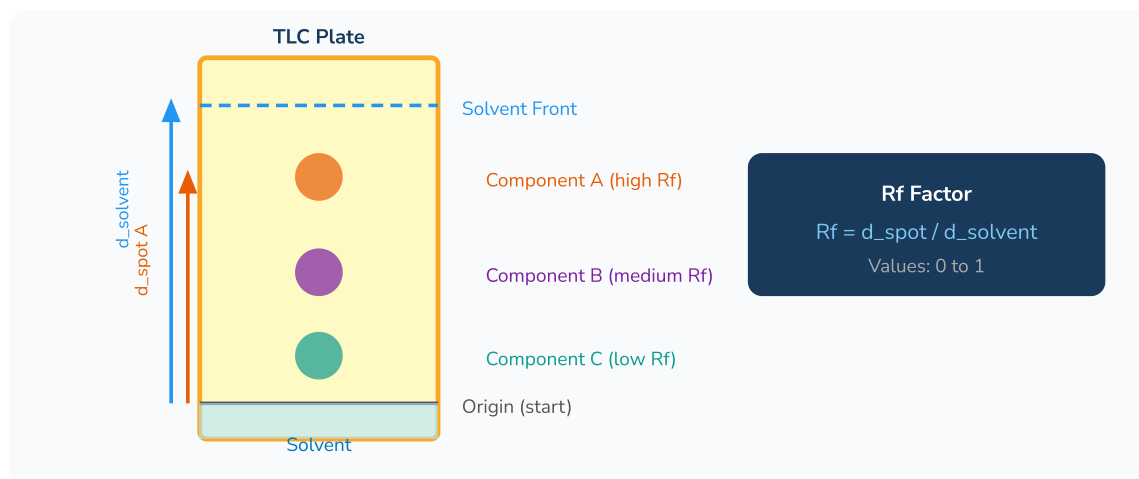


Fig 12.1 – TLC separation showing three components with different Rf values

12.5 Rf Value

$Rf = \text{Distance traveled by the solute} / \text{Distance traveled by the solvent front}$

Rf value ranges from 0 to 1

- Rf is a characteristic value for a compound under specific conditions.
- High Rf: compound has high affinity for mobile phase (less attracted to stationary phase).
- Low Rf: compound strongly retained by stationary phase.

SECTION 13

HPLC – High Performance Liquid Chromatography

13.1 Why HPLC? The Need for It

BASIC PROBLEM BEING SOLVED

Classical column chromatography is slow, gives poor resolution, and cannot separate complex mixtures efficiently. HPLC solves this by forcing the mobile phase through very fine stationary phase particles (3–10 μm) at **high pressure** (up to 400 bar), achieving much faster, highly efficient separations.

HPLC = High Performance (or High Pressure) Liquid Chromatography. It is an advanced form of liquid chromatography where the mobile phase is pumped at high pressure through a column packed with very small, uniform stationary phase particles, resulting in superior separation speed, resolution, and sensitivity.

13.2 Principle of HPLC

The separation in HPLC is based on differential interactions of sample components with:

- **Stationary phase** (e.g., silica, C18-bonded silica) — retention by adsorption, partitioning, or ion exchange
- **Mobile phase** (e.g., water/methanol, water/acetonitrile) — carries components along

Components with greater affinity for the stationary phase elute later (have longer retention times). Detection is performed by UV-Vis, fluorescence, or refractive index detectors.

13.3 HPLC Instrument – Block Diagram

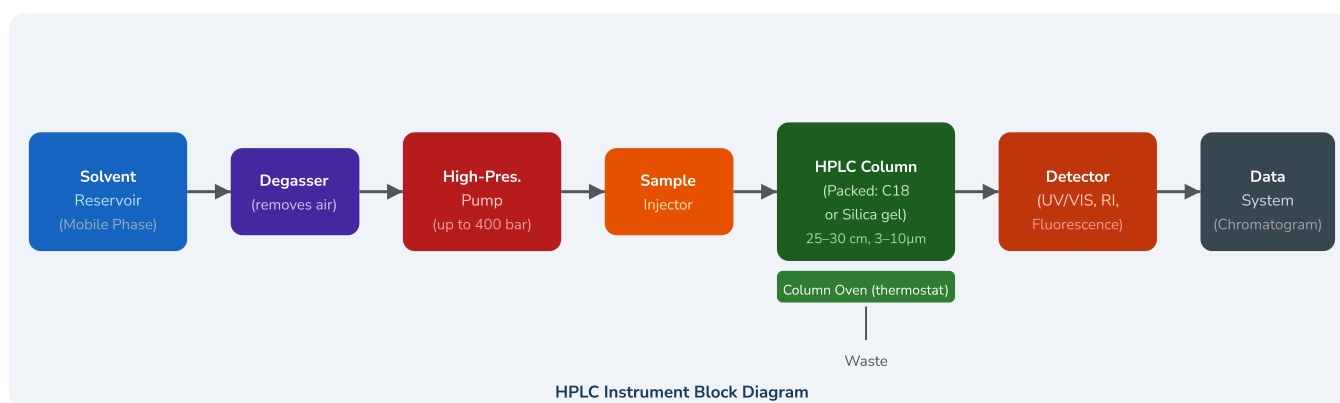


Fig 13.1 – Complete block diagram of a High Performance Liquid Chromatography (HPLC) system

13.4 Components of HPLC

Component	Function	Details
Solvent Reservoir	Holds mobile phase	Stainless steel or glass bottles; 0.5–2 L capacity

Degasser	Removes dissolved gases	Prevents bubble formation in detector; vacuum or ultrasonic degassing
High-Pressure Pump	Pushes mobile phase at constant flow	Reciprocating piston pump; pressure up to 400 bar; flow rate 0.1–10 mL/min
Sample Injector	Introduces sample into column	Manual (loop injector) or autosampler; injection volume 1–100 µL
HPLC Column	Separates sample components	Stainless steel; 10–30 cm × 4.6 mm ID; packed with 3–10 µm particles
Column Oven	Controls column temperature	Thermostat at 25–80°C for reproducibility
Detector	Detects eluted components	UV-Vis (most common), fluorescence, RI, electrochemical, MS
Data System	Records and processes signal	Software calculates peak areas, retention times; outputs chromatogram

13.5 Types of HPLC

Type	Stationary Phase	Separation Basis	Application
Normal Phase HPLC	Polar (silica)	Polarity difference; polar compounds retained more	Separating non-polar compounds
Reverse Phase HPLC (RP-HPLC)	Non-polar (C18, C8)	Hydrophobicity; non-polar retained more	Most common; drugs, peptides, vitamins
Ion Exchange HPLC	Charged resin	Ionic interactions	Amino acids, inorganic ions
Size Exclusion HPLC	Porous gel	Molecular size	Polymers, proteins

13.6 Applications of HPLC

- **Pharmaceutical industry:** Purity testing, stability studies, content uniformity of tablets and injections
- **Environmental analysis:** Detection of pesticides, herbicides, heavy metals in water/soil
- **Food industry:** Analysis of vitamins, preservatives, food additives, contaminants
- **Clinical/biomedical:** Drug monitoring in blood and urine, amino acid analysis
- **Petrochemical:** Analysis of crude oil components, gasoline additives

SECTION 14

Solved Numerical Problems

Complete step-by-step solutions — 40+ problems covering all key formulas of Unit V

Part A – Wavelength, Frequency & Energy Conversions

Problem 1: Calculate frequency of light with $\lambda = 500 \text{ nm}$

Given: $\lambda = 500 \text{ nm} = 500 \times 10^{-9} \text{ m} = 5 \times 10^{-7} \text{ m}$ | $c = 3 \times 10^8 \text{ m/s}$

- 1 Use: $c = \lambda \nu \rightarrow \nu = c/\lambda$
- 2 $\nu = (3 \times 10^8) / (5 \times 10^{-7}) = 6 \times 10^{14} \text{ Hz}$

$$\nu = 6 \times 10^{14} \text{ Hz (visible green light)}$$

Problem 2: Calculate energy of a photon with $\nu = 6 \times 10^{14} \text{ Hz}$

Given: $\nu = 6 \times 10^{14} \text{ Hz}$ | $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$

- 1 Use: $E = h\nu$
- 2 $E = 6.626 \times 10^{-34} \times 6 \times 10^{14} = 39.756 \times 10^{-20} \text{ J}$

$$E = 3.976 \times 10^{-19} \text{ J}$$

Problem 3: Convert $\lambda = 3.5 \mu\text{m}$ (IR) to wavenumber (cm^{-1})

Given: $\lambda = 3.5 \mu\text{m} = 3.5 \times 10^{-4} \text{ cm}$

- 1 Wavenumber $\tilde{\nu} = 1/\lambda$ (in cm^{-1})
- 2 $\tilde{\nu} = 1 / (3.5 \times 10^{-4}) = 2857 \text{ cm}^{-1}$

$$\tilde{\nu} = 2857 \text{ cm}^{-1} \text{ (C-H stretching region)}$$

Problem 4: Find wavelength in nm of EMR with $\nu = 3 \times 10^{15} \text{ Hz}$

Given: $\nu = 3 \times 10^{15} \text{ Hz}$ | $c = 3 \times 10^8 \text{ m/s}$

- 1 $\lambda = c/\nu = (3 \times 10^8) / (3 \times 10^{15}) = 1 \times 10^{-7} \text{ m}$
- 2 Convert: $1 \times 10^{-7} \text{ m} \times 10^9 \text{ nm/m} = 100 \text{ nm}$

$$\lambda = 100 \text{ nm (UV region)}$$

Problem 5: Calculate energy in eV for $\lambda = 400 \text{ nm}$

Given: $\lambda = 400 \text{ nm} = 4 \times 10^{-7} \text{ m}$ | $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$ | $c = 3 \times 10^8 \text{ m/s}$ | $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$

- 1 $E = hc/\lambda = (6.626 \times 10^{-34} \times 3 \times 10^8) / (4 \times 10^{-7})$
- 2 $E = 4.978 \times 10^{-19} \text{ J}$
- 3 $E \text{ in eV} = 4.978 \times 10^{-19} / 1.6 \times 10^{-19} = 3.1 \text{ eV}$

$$E = 3.1 \text{ eV (violet light at threshold of visible)}$$

Part B – Beer-Lambert Law Problems

Problem 6: Basic Beer-Lambert – Find Absorbance

Given: $\epsilon = 5000 \text{ L}\cdot\text{mol}^{-1}\cdot\text{cm}^{-1}$ | $c = 2 \times 10^{-4} \text{ mol/L}$ | $l = 1 \text{ cm}$

- 1 $A = \epsilon \times c \times l$
- 2 $A = 5000 \times 2 \times 10^{-4} \times 1 = 5000 \times 0.0002 = 1.0$

$$A = 1.0$$

Problem 7: Find %T from Absorbance

Given: $A = 1.0$

- 1 $A = -\log(T) \rightarrow \log(T) = -A = -1.0$
- 2 $T = 10^{-1} = 0.1 \rightarrow \%T = 10\%$

$$\%T = 10\% \text{ (only 10\% of light passes through)}$$

Problem 8: Find Concentration from Absorbance

Given: $A = 0.693$ | $\epsilon = 6930 \text{ L}\cdot\text{mol}^{-1}\cdot\text{cm}^{-1}$ | $l = 1 \text{ cm}$

- 1 $A = \epsilon \cdot c \cdot l \rightarrow c = A / (\epsilon \cdot l)$
- 2 $c = 0.693 / (6930 \times 1) = 0.693 / 6930 = 1 \times 10^{-4} \text{ mol/L}$

$$c = 1 \times 10^{-4} \text{ mol/L} = 0.1 \text{ mM}$$

Problem 9: Find molar absorptivity ϵ

Given: $A = 0.8$ | $c = 4 \times 10^{-3} \text{ mol/L}$ | $l = 1 \text{ cm}$

1 $\epsilon = A / (c \times l) = 0.8 / (4 \times 10^{-3} \times 1)$

2 $\epsilon = 0.8 / 0.004 = 200 \text{ L} \cdot \text{mol}^{-1} \cdot \text{cm}^{-1}$

$$\epsilon = 200 \text{ L} \cdot \text{mol}^{-1} \cdot \text{cm}^{-1} \text{ (moderate absorber)}$$

Problem 10: Find path length needed for $A = 0.5$

Given: Target $A = 0.5$ | $\epsilon = 1000 \text{ L} \cdot \text{mol}^{-1} \cdot \text{cm}^{-1}$ | $c = 5 \times 10^{-4} \text{ mol/L}$

1 $l = A / (\epsilon \times c) = 0.5 / (1000 \times 5 \times 10^{-4})$

2 $l = 0.5 / 0.5 = 1.0 \text{ cm}$

$$l = 1.0 \text{ cm (standard cuvette works!)}$$

Problem 11: %T to Absorbance conversion

Given: $\%T = 25\%$

1 $T = 25/100 = 0.25$

2 $A = -\log_{10}(T) = -\log_{10}(0.25) = -(-0.602) = 0.602$

$$A = 0.602$$

Problem 12: Absorbance to %T conversion

Given: $A = 1.5$

1 $T = 10^{-A} = 10^{-1.5} = 0.03162$

2 $\%T = 0.03162 \times 100 = 3.16\%$

$$\%T = 3.16\%$$

Problem 13: Calibration – unknown concentration from calibration data

Calibration: $A = 0.8$ when $c = 4 \times 10^{-4} \text{ mol/L}$ ($l = 1 \text{ cm}$). Unknown sample: $A = 0.5$

1 From calibration: $\epsilon = A/(c \cdot l) = 0.8/(4 \times 10^{-4} \times 1) = 2000 \text{ L} \cdot \text{mol}^{-1} \cdot \text{cm}^{-1}$

2 For unknown: $c = A/(\epsilon \cdot l) = 0.5/(2000 \times 1) = 2.5 \times 10^{-4} \text{ mol/L}$

$$c_{\text{unknown}} = 2.5 \times 10^{-4} \text{ mol/L}$$

Problem 14: Mixture – additive absorbance

Species A: $\epsilon_A = 3000$, $c_A = 2 \times 10^{-4}$ mol/L | Species B: $\epsilon_B = 5000$, $c_B = 1 \times 10^{-4}$ mol/L | $l = 1$ cm

1 $A_A = \epsilon_A \cdot c_A \cdot l = 3000 \times 2 \times 10^{-4} \times 1 = 0.6$

2 $A_B = \epsilon_B \cdot c_B \cdot l = 5000 \times 1 \times 10^{-4} \times 1 = 0.5$

3 $A_{\text{total}} = A_A + A_B = 0.6 + 0.5 = 1.1$

$A_{\text{total}} = 1.1$ (Beer's Law is additive)

Problem 15: Find dilution required for $A = 0.3$

Current $A = 1.5$ ($l = 1$ cm, $\epsilon = 15000$). Target $A = 0.3$

1 Current $c = A/(\epsilon \cdot l) = 1.5/15000 = 1 \times 10^{-4}$ mol/L

2 Target $c = 0.3/15000 = 2 \times 10^{-5}$ mol/L

3 Dilution factor = $1 \times 10^{-4} / 2 \times 10^{-5} = 5 \times$ dilution needed

Dilute 1 part sample + 4 parts solvent (5× dilution)

Part C – More Beer-Lambert & Spectroscopy Problems

Problem 16: Transmittance when $I_0 = 1000$ units, $I = 100$ units

$$I_0 = 1000, I = 100$$

- 1 $T = I/I_0 = 100/1000 = 0.1 \rightarrow \%T = 10\%$
- 2 $A = \log(I_0/I) = \log(1000/100) = \log(10) = 1$

$$A = 1.0, \%T = 10\%$$

Problem 17: Double path length effect on absorbance

$$l_1 = 1 \text{ cm}, A_1 = 0.4. \text{ Find } A \text{ when } l_2 = 2 \text{ cm (same } c \text{ and } \epsilon)$$

- 1 Since $A = \epsilon cl$, doubling l doubles A
- 2 $A_2 = 0.4 \times 2 = 0.8$

$$A_2 = 0.8 \text{ (linear relationship with path length)}$$

Problem 18: Find λ_{max} region from transition type

A compound undergoes $n \rightarrow \pi^*$ transition. Predict approximate λ_{max}

- 1 From the transition table: $n \rightarrow \pi^*$ occurs at 270–350 nm
- 2 This corresponds to near UV / visible boundary
- 3 Example: acetone shows $n \rightarrow \pi^*$ at 280 nm with $\epsilon \approx 15$

$$\lambda_{\text{max}} \approx 270\text{--}350 \text{ nm (near UV region)}$$

Problem 19: Concentration from calibration graph (slope method)

Calibration slope (A vs c) = $8000 \text{ L}\cdot\text{mol}^{-1}$ (with $l=1 \text{ cm}$). Unknown $A = 0.56$

- 1 Slope = $\epsilon \times l = 8000 \times 1 = 8000 \rightarrow \epsilon = 8000 \text{ L}\cdot\text{mol}^{-1}\cdot\text{cm}^{-1}$
- 2 $c = A/(\epsilon \cdot l) = 0.56/8000 = 7 \times 10^{-5} \text{ mol/L}$

$$c = 7 \times 10^{-5} \text{ mol/L} = 70 \text{ }\mu\text{M}$$

Problem 20: Energy comparison – which photon has more energy?

Photon A: $\lambda = 200 \text{ nm}$. Photon B: $\lambda = 600 \text{ nm}$. Compare energies.

- 1 $E \propto 1/\lambda$. Shorter $\lambda \rightarrow$ higher energy
- 2 $E_A/E_B = \lambda_B/\lambda_A = 600/200 = 3$
- 3 Photon A (UV, 200 nm) has 3 $\times$ more energy than photon B (visible, 600 nm)

$E_A = 3 \times E_B$; UV photons carry far more energy

Problem 21: Number of IR vibrational modes in SO_2

SO_2 : non-linear triatomic molecule ($N = 3$)

- 1 For non-linear: modes = $3N - 6 = 3(3) - 6 = 9 - 6 = 3$
- 2 The 3 modes are: symmetric stretch, asymmetric stretch, bending
- 3 All 3 are IR active (SO_2 is a polar molecule with dipole changes)

3 vibrational modes, all IR active

Problem 22: IR modes in CO_2 (linear molecule)

CO_2 : linear triatomic ($N = 3$)

- 1 For linear: modes = $3N - 5 = 3(3) - 5 = 4$
- 2 Modes: symmetric stretch (IR inactive), asymmetric stretch (IR active), 2 degenerate bending modes (IR active)
- 3 3 IR active modes observed; symmetric stretch is Raman active

4 total modes; 3 are IR active, 1 (sym. stretch) is IR inactive

Problem 23: Wavenumber to frequency conversion

$\tilde{\nu} = 1715 \text{ cm}^{-1}$ ($\text{C}=\text{O}$ stretch). Find frequency in Hz.

- 1 $\lambda = 1/\tilde{\nu} = 1/1715 \text{ cm} = 5.83 \times 10^{-4} \text{ cm} = 5.83 \times 10^{-6} \text{ m}$
- 2 $\nu = c/\lambda = (3 \times 10^8) / (5.83 \times 10^{-6}) = 5.15 \times 10^{13} \text{ Hz}$

$\nu = 5.15 \times 10^{13} \text{ Hz}$

Problem 24: Calculate R_f value in TLC

Spot traveled 4.5 cm. Solvent front: 9 cm from origin.

- 1 $R_f = \text{Distance by spot} / \text{Distance by solvent} = 4.5 / 9.0$

2 $R_f = 0.50$

$R_f = 0.50$ (midpoint; moderate affinity for stationary phase)

Problem 25: Identify component from R_f value

Known: caffeine $R_f = 0.45$; aspirin $R_f = 0.72$; paracetamol $R_f = 0.32$ (under specific TLC conditions). Unknown sample shows $R_f = 0.73$.

- 1 Match $R_f = 0.73$ with known values
- 2 Aspirin $R_f = 0.72 \approx 0.73 \rightarrow$ best match

Unknown is likely Aspirin ($R_f \approx 0.72$)

Problem 26: Hooke's Law – compare C–H vs C–D stretching frequency

C–H: μ_{CH} . C–D: μ_{CD} . If H is replaced by D (mass doubles), how does ν change?

- 1 $\nu \propto \sqrt{k/\mu}$. For C–H: $\mu = (12 \times 1)/(12+1) = 0.923$ amu
- 2 For C–D: $\mu = (12 \times 2)/(12+2) = 1.714$ amu
- 3 $\nu_{CD}/\nu_{CH} = \sqrt{\mu_{CH}/\mu_{CD}} = \sqrt{0.923/1.714} = \sqrt{0.5385} = 0.734$
- 4 C–H at $\sim 3000 \text{ cm}^{-1} \rightarrow$ C–D at $\sim 3000 \times 0.734 = \sim 2200 \text{ cm}^{-1}$

C–D stretch appears $\sim 2200 \text{ cm}^{-1}$ (lower than C–H due to heavier D)

Problem 27: Two-cell absorbance comparison

Cell 1: $l = 0.5 \text{ cm}$, $A = 0.3$. Cell 2: $l = 2 \text{ cm}$. Same solution. Find A_2 .

- 1 Since $A = \epsilon cl$: $A_1/A_2 = l_1/l_2$
- 2 $A_2 = A_1 \times (l_2/l_1) = 0.3 \times (2/0.5) = 0.3 \times 4 = 1.2$

$A_2 = 1.2$

Problem 28: Find concentration from %T measurement

%T = 63.1% | $\epsilon = 1000 \text{ L} \cdot \text{mol}^{-1} \cdot \text{cm}^{-1}$ | $l = 1 \text{ cm}$

- 1 $T = 63.1/100 = 0.631$
- 2 $A = -\log(0.631) = -(-0.2) = 0.2$
- 3 $c = A/(\epsilon \cdot l) = 0.2/(1000 \times 1) = 2 \times 10^{-4} \text{ mol/L}$

$c = 2 \times 10^{-4} \text{ mol/L}$

Problem 29: Beer-Lambert with unit conversion (mg/L to mol/L)

$A = 0.5$ | $\epsilon = 10,000 \text{ L}\cdot\text{mol}^{-1}\cdot\text{cm}^{-1}$ | $l = 1 \text{ cm}$ | MW of compound = 200 g/mol

- 1 $c \text{ (mol/L)} = A/(\epsilon \cdot l) = 0.5/10000 = 5 \times 10^{-5} \text{ mol/L}$
- 2 $c \text{ (mg/L)} = c \text{ (mol/L)} \times \text{MW} \times 1000 = 5 \times 10^{-5} \times 200 \times 1000 = 10 \text{ mg/L}$

$$c = 5 \times 10^{-5} \text{ mol/L} = 10 \text{ mg/L}$$

Problem 30: Minimum detectable concentration (detection limit)

Minimum detectable $A = 0.001$ | $\epsilon = 100,000 \text{ L}\cdot\text{mol}^{-1}\cdot\text{cm}^{-1}$ | $l = 1 \text{ cm}$

- 1 $c_{\text{min}} = A_{\text{min}} / (\epsilon \cdot l) = 0.001 / (100,000 \times 1)$
- 2 $c_{\text{min}} = 1 \times 10^{-8} \text{ mol/L} = 10 \text{ nM}$

Detection limit = 10 nM (excellent sensitivity – typical of fluorescent compounds)

Part D – Advanced Problems

Problem 31: Determine if mixture obeys Beer-Lambert Law

At $c = 0.001 \text{ M}$: $A = 0.2$. At $c = 0.002 \text{ M}$: $A = 0.4$. At $c = 0.01 \text{ M}$: $A = 1.8$ (not 2.0)

- 1 For dilute range (0.001–0.002 M): $\Delta A/\Delta c = (0.4 - 0.2)/(0.001) = 200 \text{ L/mol}$ — linear ✓
- 2 At high c (0.01 M): expected $A = 200 \times 0.01 = 2.0$ but got 1.8 — deviation
- 3 This is a negative deviation from Beer-Lambert Law (common at high concentration)

Beer-Lambert Law is obeyed at low concentrations ($c < 0.01 \text{ M}$); positive deviation at high c

Problem 32: Absorption spectrum – identify highest energy transition

Compound has peaks at: 250 nm ($\epsilon = 18000$), 310 nm ($\epsilon = 50$), 370 nm ($\epsilon = 8$)

- 1 Higher energy → shorter wavelength. 250 nm is shortest wavelength → highest energy
- 2 250 nm with high $\epsilon = \pi \rightarrow \pi^*$ transition (strong absorber)
- 3 310–370 nm with low $\epsilon = n \rightarrow \pi^*$ transitions (weak absorbers, lone pair involved)

250 nm is highest energy transition ($\pi \rightarrow \pi^*$); 310/370 nm are $n \rightarrow \pi^*$ transitions

Problem 33: Convert frequency to energy in kJ/mol

$\nu = 3 \times 10^{14}$ Hz. Find energy in kJ/mol.

- 1 $E \text{ per photon} = h\nu = 6.626 \times 10^{-34} \times 3 \times 10^{14} = 1.99 \times 10^{-19} \text{ J}$
- 2 Per mole: $E = 1.99 \times 10^{-19} \times 6.022 \times 10^{23} = 1.198 \times 10^5 \text{ J/mol}$
- 3 $E = 1.198 \times 10^5 / 1000 = 119.8 \text{ kJ/mol}$

$$E = 119.8 \text{ kJ/mol} \approx 120 \text{ kJ/mol}$$

Problem 34: HPLC – Identify peak from retention time

HPLC chromatogram shows peaks at $t^R = 2.1, 4.5, 7.3$ min. Standard: A at 2.1 min, B at 4.5 min, C at 7.3 min. Which component is retained most?

- 1 Longer retention time = component interacts more strongly with stationary phase
- 2 C at 7.3 min has longest $t^R \rightarrow$ most retained by stationary phase
- 3 A at 2.1 min is least retained $\rightarrow$ has highest affinity for mobile phase

Component C is most retained ($t^R = 7.3$ min); component A elutes first

Problem 35: Absorption of 1710 cm^{-1} in IR — assign functional group

IR spectrum shows a strong absorption at 1710 cm^{-1} . Identify the functional group.

- 1 Reference: C=O stretch (aldehyde/ketone) appears at $1680\text{--}1780 \text{ cm}^{-1}$
- 2 1710 cm^{-1} is within the carbonyl range
- 3 Ketone C=O typically at $1705\text{--}1725 \text{ cm}^{-1}$; aldehyde C=O at $1720\text{--}1740 \text{ cm}^{-1}$

The band at 1710 cm^{-1} indicates a C=O stretch – most likely a ketone carbonyl group

Problem 36: Find ϵ from published data

$A = 0.312$ | $c = 1.5 \times 10^{-4} \text{ M}$ | $l = 2 \text{ cm}$. Find ϵ .

- 1 $\epsilon = A/(c \times l) = 0.312 / (1.5 \times 10^{-4} \times 2)$
- 2 $\epsilon = 0.312 / 3 \times 10^{-4} = 1040 \text{ L}\cdot\text{mol}^{-1}\cdot\text{cm}^{-1}$

$$\epsilon = 1040 \text{ L}\cdot\text{mol}^{-1}\cdot\text{cm}^{-1}$$

Problem 37: Quantify unknown using standard comparison

Standard: 50 $\mu\text{g/mL}$ solution shows $A_{\text{std}} = 0.400$. Unknown shows $A_{\text{unk}} = 0.250$ (same cuvette, same λ).

- 1 Beer-Lambert: $A = \epsilon cl$, so $A \propto c$ (if ϵ , l constant)
- 2 $c_{\text{unk}} / c_{\text{std}} = A_{\text{unk}} / A_{\text{std}} = 0.250 / 0.400 = 0.625$
- 3 $c_{\text{unk}} = 0.625 \times 50 \mu\text{g/mL} = 31.25 \mu\text{g/mL}$

Concentration of unknown = 31.25 $\mu\text{g/mL}$

Problem 38: Calculate % Transmittance for two stacked cuvettes

One cuvette ($l = 1 \text{ cm}$) gives $\%T = 50\%$. Two identical cuvettes in series ($l = 2 \text{ cm}$ total). Find $\%T$.

- 1 $\%T = 50\% \rightarrow T = 0.5 \rightarrow A = -\log(0.5) = 0.301$ (for 1 cm)
- 2 For 2 cm: $A_2 = 2 \times 0.301 = 0.602$
- 3 $T_2 = 10^{-0.602} = 0.25 \rightarrow \%T_2 = 25\%$

$\%T = 25\%$ (not 0%; $\%T$ squares: $50\% \times 50\% = 25\%$)

Problem 39: Frequency of C=O group in IR from Hooke's Law

$k (\text{C=O}) \approx 1200 \text{ N/m}$. μ for C=O = $(12 \times 16) / (12 + 16) \text{ amu} = 6.857 \text{ amu} = 6.857 \times 1.66 \times 10^{-27} \text{ kg} = 1.138 \times 10^{-26} \text{ kg}$

- 1 $\nu = (1/2\pi) \sqrt{k/\mu} = (1/2\pi) \times \sqrt{1200/1.138 \times 10^{-26}}$
- 2 $k/\mu = 1200/1.138 \times 10^{-26} = 1.054 \times 10^{29}$
- 3 $\sqrt{1.054 \times 10^{29}} = 1.027 \times 10^{14.5} \approx 3.247 \times 10^{13}$
- 4 $\nu = 3.247 \times 10^{13} / (2\pi) = 5.17 \times 10^{12} \text{ Hz} \rightarrow \tilde{\nu} = \nu/c = 5.17 \times 10^{12} / (3 \times 10^{10}) \approx 1724 \text{ cm}^{-1}$

$\tilde{\nu} \approx 1720 \text{ cm}^{-1}$ ✓ (matches experimental C=O at 1680–1780 cm^{-1})

Problem 40: Identify chromophore from UV data

Compound shows: $\lambda_{\text{max}} = 217 \text{ nm}$, $\epsilon = 21,000$. Identify the likely chromophore.

- 1 217 nm and $\epsilon \approx 21,000$ matches $\pi \rightarrow \pi^*$ transition in a diene system
- 2 Conjugated diene (butadiene type) absorbs at $\sim 217 \text{ nm}$ with $\epsilon \approx 21,000$
- 3 Simple alkenes absorb at 170–180 nm, so this is an extended conjugated system

Chromophore: conjugated diene ($-\text{C}=\text{C}-\text{C}=\text{C}-$); $\pi \rightarrow \pi^*$ transition at 217 nm

Problem 41: Absorbance at different wavelengths

At $\lambda_1 = 430 \text{ nm}$: $\epsilon_1 = 12000$. At $\lambda_2 = 530 \text{ nm}$: $\epsilon_2 = 3000$. $c = 5 \times 10^{-5} \text{ M}$, $l = 1 \text{ cm}$.
Find A at each λ .

1 $A_1 = \epsilon_1 c l = 12000 \times 5 \times 10^{-5} \times 1 = 0.6$

2 $A_2 = \epsilon_2 c l = 3000 \times 5 \times 10^{-5} \times 1 = 0.15$

A at 430 nm = 0.6; A at 530 nm = 0.15 (stronger absorption at 430 nm)

Problem 42: HPLC resolution calculation

Two peaks: Peak 1 at $t^R_1 = 5.0 \text{ min}$, $w_1 = 0.4 \text{ min}$. Peak 2 at $t^R_2 = 5.8 \text{ min}$, $w_2 = 0.5 \text{ min}$.

1 Resolution $R = 2(t^R_2 - t^R_1) / (w_1 + w_2)$

2 $R = 2(5.8 - 5.0) / (0.4 + 0.5) = 2 \times 0.8 / 0.9 = 1.6 / 0.9 = 1.78$

3 $R > 1.5$ means baseline separation (good resolution)

$R = 1.78 \rightarrow$ Good baseline resolution between the two peaks

SECTION 15

Questions & Answers — 2M and 5M

Part A – 2-Mark Questions & Answers

Q1. Define Beer-Lambert Law. 2M

Ans: Beer-Lambert Law states that the absorbance (A) of a solution is directly proportional to the molar absorptivity (ϵ), concentration (c), and path length (l): $A = \epsilon cl$. It forms the basis for quantitative spectrophotometric analysis.

Q2. What is absorbance? How is it related to transmittance? 2M

Ans: Absorbance (A) = $\log_{10}(I_0/I) = -\log_{10}(T)$, where T is transmittance. If T = 0.1 (10%), then A = 1. Higher absorbance means less light passes through.

Q3. Define chromophore. Give two examples. 2M

Ans: A chromophore is any unsaturated group in a molecule responsible for UV-Vis absorption. Examples: C=C (alkene) absorbs at 170 nm; C=O (carbonyl) absorbs at ~280 nm ($n \rightarrow \pi^*$).

Q4. What is an auxochrome? How does it differ from a chromophore? 2M

Ans: An auxochrome is a functional group (e.g., -OH, -NH₂) that does not absorb UV-Vis alone but shifts the absorption of a chromophore to longer wavelengths (bathochromic shift) when attached to it. Chromophores absorb independently; auxochromes modify chromophore absorption.

Q5. State the selection rule for IR spectroscopy. 2M

Ans: A molecular vibration is IR active only if there is a change in the **dipole moment (μ)** of the molecule during that vibration. Symmetrical vibrations with no dipole change are IR inactive.

Q6. What is the fingerprint region in IR spectroscopy? 2M

Ans: The fingerprint region (1500–400 cm⁻¹) contains complex, overlapping absorption bands unique to each compound. It is used for identification by comparison with reference spectra, like a molecular fingerprint.

Q7. Define R_f value in chromatography. What is its range? 2M

Ans: R_f (retardation factor) = Distance traveled by compound / Distance traveled by solvent front. It ranges from 0 to 1 and is a characteristic value for a compound under specific TLC conditions.

Q8. What are the advantages of HPLC over classical column chromatography? 2M

Ans: HPLC advantages: (1) Much faster analysis (minutes vs hours), (2) Better resolution due to fine 3–10 μ m particles, (3) Higher sensitivity and reproducibility, (4) Automated injection and data recording, (5) Suitable for thermally labile compounds.

Q9. Write the relationship between wavelength, frequency and speed of light. 2M

Ans: $c = \lambda \nu$, where c = speed of light (3 × 10⁸ m/s), λ = wavelength (m), ν = frequency (Hz). This is the fundamental wave equation for electromagnetic radiation.

Q10. Why is glass not used as sample cell in UV spectroscopy? 2M

Ans: Glass strongly absorbs UV radiation below 350 nm, preventing UV light from reaching the sample. **Quartz (fused silica)** cells are used instead, as quartz is transparent to UV down to about 190 nm.

Q11. What is a bathochromic shift? 2M

Ans: A bathochromic shift (red shift) is the shift of λ_{max} towards **longer wavelengths** (lower energy). It is caused by conjugation, auxochromes, or change in solvent polarity.

Q12. State the number of IR vibrational modes for water (H₂O). 2M

Ans: Water is a non-linear molecule with $N = 3$ atoms. $\text{Modes} = 3N - 6 = 9 - 6 = 3$ **modes:** (1) symmetric O–H stretch, (2) asymmetric O–H stretch, (3) H–O–H bending (scissoring). All three are IR active.

Part B – 5-Mark Questions & Answers

Q1. Explain Beer-Lambert Law and its limitations. 5M

Beer-Lambert Law: $A = \epsilon cl$, where A = absorbance, ϵ = molar absorptivity, c = concentration, l = path length.

- Lambert's Law: $A \propto \text{path length } l$ (at constant c)
- Beer's Law: $A \propto \text{concentration } c$ (at constant l)
- Combined: $A = \epsilon cl$
- Graphically: A vs c is linear with slope $= \epsilon l$

Limitations:

- **Chemical deviations:** At high concentrations (>0.01 M), solute–solute interactions change ϵ . Chemical equilibria (dissociation, dimerization) alter absorbing species.
- **Instrumental deviations:** Non-monochromatic light: Beer's law holds strictly only for monochromatic radiation. Stray light causes negative deviations at high absorbance.
- **Physical deviations:** Fluorescence from sample, scattering by colloidal particles, change in refractive index at high concentration.

Practical note: The law is most reliable in the 0.1–1.0 absorbance range (10–80%T).

Q2. Describe the instrumentation of a UV-Visible spectrophotometer with a block diagram. 5M

A UV-Vis spectrophotometer consists of:

- **Radiation Source:** Deuterium (D₂) lamp for UV (190–400 nm), tungsten-halogen for visible (350–700 nm).
- **Monochromator:** Prism or diffraction grating isolates specific wavelengths.
- **Beam Splitter:** Divides light into sample and reference beams (double-beam design).
- **Sample and Reference Cells:** Quartz cuvettes (1 cm path length); reference holds blank solvent.
- **Detector:** Photomultiplier tube (PMT) converts light to electrical signal.
- **Amplifier/Recorder:** Amplifies signal and plots spectrum (A vs λ).

Working: Light from source passes through monochromator → splits → sample and reference → detector computes ratio I/I_0 → absorbance $A = \log(I_0/I)$ plotted vs λ .

[Draw block diagram as in Fig 8.1]

Q3. Explain the principle and instrumentation of HPLC with a block diagram. 5M

Principle: HPLC separates components based on differential partitioning between a stationary phase (e.g., C18 bonded silica) and a mobile phase (e.g., water/methanol), pumped at high pressure (up to 400 bar). Components with greater affinity for the stationary phase elute later.

Components:

- Solvent reservoir → Degasser → High-pressure pump → Injector → Column (with thermostat) → Detector → Data system
- High-pressure pump: ensures constant flow rate (0.1–10 mL/min)
- Column: stainless steel, 10–30 cm long, 3–10 μm particle size
- Detector: UV-Vis most common; fluorescence and RI detectors also used

Applications: Pharmaceutical purity testing, food additives, environmental pollutants, clinical analysis.

[Draw block diagram as in Fig 13.1]

Q4. Discuss the types of electronic transitions in UV-Vis spectroscopy and give examples. 5M

Molecules contain electrons in σ , π , and n (non-bonding) orbitals. UV-Vis absorption causes electrons to be promoted to higher-energy antibonding (σ^* , π^*) orbitals.

Types of transitions:

1. $\sigma \rightarrow \sigma^*$: Requires very high energy (<200 nm, vacuum UV). Only alkanes (C–C, C–H). Not observed in routine UV-Vis. Example: CH_4 at 125 nm.
2. $n \rightarrow \sigma^*$: 150–250 nm, moderate ϵ . Requires heteroatom lone pairs. Example: methanol (184 nm), trimethylamine.
3. $\pi \rightarrow \pi^*$: 170–250 nm, very high ϵ (10,000–100,000). Requires C=C, C=O, aromatic rings. Example: benzene (254 nm), butadiene (217 nm).
4. $n \rightarrow \pi^*$: 270–350 nm, very low ϵ (10–100). Requires C=O, C=N with lone pair. Example: acetone (280 nm, $\epsilon \approx 15$).

Energy order: $n \rightarrow \pi^* < \pi \rightarrow \pi^* < n \rightarrow \sigma^* < \sigma \rightarrow \sigma^*$

Q5. Explain the fundamental vibrational modes in IR spectroscopy and give the selection rule. 5M

Molecular vibrations are of two types:

1. Stretching Vibrations (bond length changes):

- Symmetric stretch: both bonds extend/compress simultaneously (may be IR inactive)
- Asymmetric stretch: one extends while other compresses (IR active)

2. Bending Vibrations (bond angle changes):

- In-plane: Scissoring (angle decreases), Rocking (both atoms move same direction)
- Out-of-plane: Wagging (both move out of plane together), Twisting (move in opposite out-of-plane directions)

Number of modes: Non-linear: $3N-6$; Linear: $3N-5$ (N = number of atoms).

Selection Rule: A vibration is IR active only if it causes a change in dipole moment ($\Delta\mu \neq 0$). Symmetric molecules like H_2 and homonuclear diatomics are IR inactive.

[Draw Fig 10.1 showing all 6 modes]

SECTION 16

⚡ Quick Revision Sheet

All essential formulas, key concepts, and exam tricks — for last-minute preparation

Key Formulas

MASTER FORMULA COLLECTION – UNIT V

$$c = \lambda \nu \quad (\text{Wave eqn}) \quad E = h\nu = hc/\lambda \quad (\text{Planck})$$

$$\tilde{\nu} = 1/\lambda \quad (\text{cm}^{-1}) \quad (\text{Wavenumber}) \quad A = \epsilon c l \quad (\text{Beer-Lambert})$$

$$A = \log(I_0/I) = -\log(T) \quad \%T = (I/I_0) \times 100$$

$$\nu_{\text{vib}} = (1/2\pi)\sqrt{k/\mu} \quad (\text{Hooke's Law})$$

$$\mu = m_1 m_2 / (m_1 + m_2) \quad (\text{Reduced mass})$$

$$Rf = d_{\text{spot}} / d_{\text{solvent}} \quad R = 2(t^R_2 - t^R_1) / (w_1 + w_2) \quad (\text{HPLC Resolution})$$

ELECTROMAGNETIC SPECTRUM

- γ -rays: <0.01 nm (highest energy)
- X-rays: 0.01–10 nm
- UV: 10–400 nm
- Visible: 400–700 nm (ROYGBIV)
- IR: 700 nm – 1 mm
- Microwave: 1 mm – 1 m
- Radio: >1 m (lowest energy)

ELECTRONIC TRANSITIONS

- $\sigma \rightarrow \sigma^*$: <200 nm, very high energy
- $n \rightarrow \sigma^*$: 150–250 nm
- $\pi \rightarrow \pi^*$: 170–250 nm, high ϵ
- $n \rightarrow \pi^*$: 270–350 nm, low ϵ
- Higher ϵ = stronger absorber
- Conjugation $\rightarrow$ red shift (batho)
- Auxochromes: $-\text{OH}$, $-\text{NH}_2$

IR KEY ABSORPTIONS

- O–H free: 3580–3650 cm^{-1}
- N–H: 3300–3500 cm^{-1}
- C–H: 2850–3000 cm^{-1}
- $\text{C}\equiv\text{N}$ / $\text{C}\equiv\text{C}$: 2100–2260 cm^{-1}
- C=O: 1680–1780 cm^{-1} ★
- C=C (arom): 1500–1600 cm^{-1}

BEER-LAMBERT TIPS

- $A = \epsilon c l$ (all in consistent units!)
- l usually = 1 cm standard
- A is additive for mixtures
- Deviations: high conc., stray light
- Optimal range: $A = 0.1$ – 1.0
- $\%T$ and A are inversely related

- Fingerprint: 400–1500 cm^{-1}

- $A = 0 \rightarrow 100\%T$; $A = 1 \rightarrow 10\%T$

CHROMATOGRAPHY TERMS

- $R_f = 0$ to 1 (dimensionless)
- High R_f = mobile phase preference
- TLC: silica on plate (polar)
- RP-HPLC: C18 (non-polar) most common
- Longer t_R = more retained
- $R > 1.5$ = baseline resolution

HPLC VS CLASSICAL LC

- HPLC pressure: up to 400 bar
- Particle size: 3–10 μm
- Analysis time: minutes
- Pump: reciprocating piston
- Detector: UV-Vis / RI / fluores.
- Column: stainless steel, 10–30 cm
- Degasser removes dissolved air

Comparison Tables

Parameter	UV-Vis Spectroscopy	IR Spectroscopy	HPLC
Region	200–800 nm	4000–400 cm^{-1}	—
Transition type	Electronic	Vibrational	—
Main use	Quantitative analysis	Structure/ID	Separation/Quant.
Cell material	Quartz/glass	KBr/NaCl	Stainless steel column
Key law/principle	Beer-Lambert	Hooke's Law	Partition/adsorption
Detection limit	nM range (high ϵ)	Moderate	ppm–ppb

Memory Tricks & Short Cuts

EXAM TRICKS

- **"A= ϵ cl" mnemonic:** "Every Chemistry Lab" (E-C-L)
- **Energy order:** $n \rightarrow \pi^*$ (weakest) < $\pi \rightarrow \pi^*$ < $n \rightarrow \sigma^*$ < $\sigma \rightarrow \sigma^*$ (strongest)
- **When A = 1:** %T = 10%. When A = 2: %T = 1%.
- **IR rule:** If the vibration changes dipole $\rightarrow$ IR active. Symmetric = usually IR inactive.
- **C=O absorption $\sim 1715 \text{ cm}^{-1}$** is THE most diagnostic IR band.
- **HPLC 3 P's:** Pressure, Particles (small), Performance.
- **R_f values:** R_f close to 1 = travels far = more like mobile phase.
- **UV-Vis cell:** Quartz for UV; Glass for visible only.
- **Wavenumber shortcut:** $\tilde{\nu} (\text{cm}^{-1}) = 10^4 / \lambda(\mu\text{m})$

Important Constants & Values

$$h \text{ (Planck's constant)} = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$$

$$c \text{ (speed of light)} = 3 \times 10^8 \text{ m/s} = 3 \times 10^{10} \text{ cm/s}$$

$$N_A \text{ (Avogadro's number)} = 6.022 \times 10^{23} \text{ mol}^{-1}$$

$$1 \text{ nm} = 10^{-9} \text{ m} \quad | \quad 1 \text{ }\mu\text{m} = 10^{-6} \text{ m} \quad | \quad 1 \text{ cm}^{-1} = 100 \text{ m}^{-1}$$

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J} \quad | \quad 1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$$

Final Exam Strategy:

- For 2M questions: always give a definition + formula or example.
- For 5M questions: define + principle + labelled block diagram + applications (at least 3).
- Numerical problems: always write the formula first, then substitute values with units.
- Diagrams for IR and UV-Vis: practise drawing them in <3 minutes.

✦ Engineering Chemistry – Unit V – Instrumental Methods and Applications ✦

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